Car Insurance Portfolio Analysis

Business Intelligence Internship Assignment Report

Prepared by: Hritik Mishra

Executive Summary

This analytical study examines a simulated car insurance portfolio combining Python-based data simulation, MySQL analytical queries, and Power BI dashboard visualizations. The analysis evaluates profitability and risk patterns across 1 million policy records.

Key metrics include premium collection, claim costs, loss ratio, and claim trends. Results reveal a loss ratio exceeding 1, indicating claim payouts surpass collected premiums.

Project Objectives

Data Simulation

Generate large-scale policy data with realistic distributions

Claim Simulation

Simulate claim events based on defined business rules

Performance Analysis

Evaluate claim costs relative to premiums collected

Risk Patterns

Identify risk patterns across policy tenures and time periods

Business Context

Insurance companies collect premiums from customers in exchange for financial protection against potential losses. Profitability depends on balancing premium income with claim payouts.

By analyzing policy data and claims history, insurers can identify trends, assess risk exposure, and adjust pricing strategies. This simulated scenario evaluates how claim events affect portfolio profitability and risk distribution.

Policy Sales Dataset

1M

Policy Records

Simulated policies purchased in 2024

₹100K

Vehicle Value

Uniform value for all policies

₹100

Annual Premium

Per year of policy tenure

Policy Tenure Distribution

- 20% - 1 year policies
- 30% - 2 year policies
- 40% - 3 year policies
- 10% - 4 year policies

Timeline

Purchase Period: January 1, 2024 – December 31, 2024

Policy Start: Purchase date + 365 days

Claims Dataset

2025 Claims

30% of vehicles purchased on 7th, 14th, 21st, 28th of each month file claims

Claim Amount

₹10,000 10% of vehicle value)

2026 Additional

10% of 4-year policies file additional claim January-February 2026

Tools and Technologies

Python

Pandas and NumPy for dataset simulation and preprocessing

MySQL

Analytical queries and data aggregation

Power BI

Interactive dashboard visualization

SQL

Calculating metrics: loss ratio, claim cost, earned premium

Key Metrics

2.05

Loss Ratio

Claim payouts exceed collected premium

49K

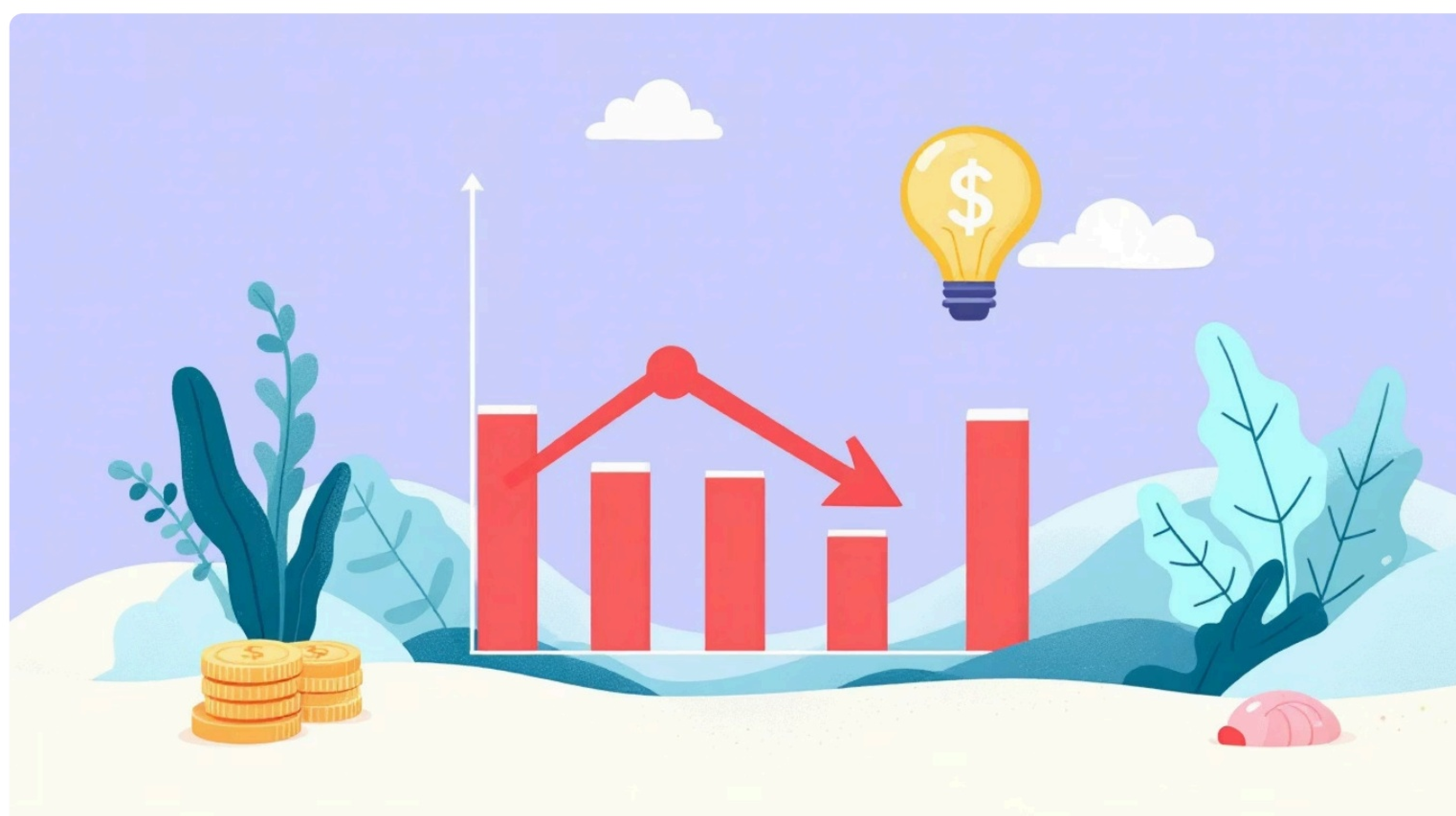
Claim Count

Total claims filed across portfolio

Financial Summary

Total Premium Collected: ₹240,000,000

Total Claim Cost: ₹491,850,000



Key Insights

Loss Ratio Exceeds 1

Claim payouts are significantly higher than collected premium, indicating unsustainable pricing model

Three-Year Policies Most Profitable

Demonstrate lowest loss ratio, representing optimal balance between coverage duration and risk exposure

2026 Claim Increase

Claim activity stable in 2025 but increases in early 2026 due to additional claims from long-tenure policies

Longer Tenures = Higher Risk

Extended coverage duration increases exposure to claim events and overall portfolio risk

Conclusion

This project demonstrates how Business Intelligence techniques can analyze insurance portfolios and evaluate financial performance. By combining data simulation, SQL analysis, and dashboard visualization, valuable insights emerge into risk patterns and profitability drivers.

The analysis highlights the importance of carefully balancing premium pricing with claim exposure to maintain a sustainable insurance portfolio. Three-year policies represent the most profitable segment, while longer tenures increase risk exposure significantly.